LENDING GUARD

Credit Portfolio Quality & Loss Intelligence

End-to-End BFSI Data Analytics Project Report

Python & Pandas • SQL Server • Power BI

PROJECT PURPOSE
Investigate where loan delinquency and defaults are concentrated, examine customer, loan and acquisition characteristics associated with risk, and understand how credit losses change over time.

Prepared as a portfolio project using synthetic lending data

Executive Summary

Lending Guard is an end-to-end lending analytics project designed to demonstrate how a Data Analyst can translate multi-table loan data into risk-management insights and stakeholder-facing reporting. The workflow moves from business problem definition and initial validation to data preparation in Python/Pandas, business analysis in SQL Server, and interactive reporting in Power BI.

The current dashboard reports 120.25K applications, 77.03K loans, 3,656 delinquent loans, 2,989 defaulted loans and 903 written-off loans. The SQL cohort analysis reported a lower default rate in the 2025 disbursement cohort than in 2024 (3.75% versus 4.33%) and a lower written-off rate (1.13% versus 1.30%). The dashboard also highlights differences by acquisition channel, employment status and loan type. These are associations that warrant investigation, not proof of causation.

KEY TAKEAWAY
Portfolio quality should not be assessed using default rate alone. A lender should examine risk concentrations, cohort maturity, exposure, credit losses and recoveries together. Data-quality issues in the synthetic source tables also need to be disclosed.

Project at a Glance

Item	Description
Domain	Banking, Financial Services and Insurance (BFSI) / Lending
Business focus	Loan portfolio quality, delinquency, defaults, write-offs and credit losses
Tools	Python, Pandas, SQL Server and Power BI
Data	Synthetic multi-table lending dataset supplied as CSV files
Reporting layer	Two-page interactive Power BI dashboard
Primary audience	Credit-risk, lending operations, acquisition and collections stakeholders

1. Business Problem and Objective

A lending business must grow its loan portfolio while managing credit risk. Portfolio growth alone does not show whether lending is healthy: growth can be accompanied by delinquency, defaults, write-offs, collection costs and financial losses. Aggregate portfolio metrics can also conceal meaningful differences across customer segments, loan products, underwriting policy versions and acquisition channels.

The project aims to answer the following overarching question:

BUSINESS PROBLEM
Where are delinquency and defaults concentrated, which observable customer, loan and acquisition characteristics are associated with those outcomes, and what do the trends indicate about the portfolio's financial risk?

Objectives:

- Measure portfolio scale and the distribution of loan outcomes.
- Compare delinquency and default rates across customer, loan, underwriting and acquisition segments.
- Track default, delinquency, write-off and credit-loss trends over time.
- Validate whether the available risk indicators align with observed loan outcomes.
- Translate findings into targeted investigations and measurable business recommendations.

2. Dataset Overview

The project uses a synthetic lending dataset distributed across multiple CSV tables. The tables represent applications, loans, customers, credit-risk assessments, repayments, collections, acquisition activity and monthly business KPIs. The repayments file was confirmed to contain 489,438 data rows excluding the header.

Table	Key fields / contents	Purpose
applications	Application/customer IDs, application attributes, channel and status	Demand, approvals and conversion

loans	Loan/application/customer IDs, disbursement date, amount, term, rate, loan type, risk band, channel, policy version, maturity and status	Core portfolio and loan outcomes
customers	Customer ID, age, gender, city, employment, tenure, income and other attributes	Customer-segment comparisons
credit_risk / history	Credit score, probability of default, DTI, previous payment history and risk attributes	Risk-indicator validation and segmentation
collections	Loan ID, delinquency stage, dates, recovery status, recovered amount and costs	Delinquency and collection outcomes
repayments	Repayment/loan IDs, due/payment dates, scheduled/actual payment, DPD and payment flags	Payment-level checks and discrepancy review
acquisition	Customer ID, acquisition date/channel, campaign, partner, cost and conversion	Channel performance and acquisition risk
monthly_business_kpis	Monthly applications, approvals, disbursements, portfolio, default rate, loss, recovery, revenue and costs	Aggregate monthly performance and reconciliation

Data Relationships and Grain

The loans table is the core of the loan-outcome analysis. It is related to customers and credit-risk information, while applications provide funnel context and collections/repayments provide event-level context. A single loan may have multiple repayment, collection or risk-history records. Joins therefore need to respect table grain; otherwise, one-to-many relationships can multiply rows and inflate loan counts. Distinct loan IDs and an appropriately selected loan-level risk record should be used where applicable.

3. Initial Validation of the Business Problem

Initial exploration established that the dataset contained multiple loan statuses, delinquency/default outcomes, risk-related attributes and monthly financial measures. This was sufficient to support an investigation of portfolio quality and financial impact, while leaving the underlying drivers to be tested.

- Portfolio validation: inspect the distribution of paid, active, defaulted, restructured and written-off loans.
- Risk-segment validation: confirm that fields such as loan type, employment status, risk band, underwriting policy and acquisition channel are available for comparison.
- Financial-impact validation: confirm the presence of credit-loss and recovery measures for investigating monetary outcomes.
- Time-series validation: examine whether disbursement dates and monthly KPI fields support meaningful trend and cohort analysis.

INITIAL CONCLUSION

The data supports a coherent analysis of loan portfolio quality, risk segmentation and financial loss. It does not establish the causes of default in advance; those must be investigated, and source inconsistencies must be documented.

4. Hypotheses and Investigation Plan

Hypothesis	Test / comparison
H1. Risk differs across customer groups.	Compare delinquency and default rates by employment status, income, job sector, gender and other available attributes.
H2. Some loan products carry higher risk.	Compare outcomes across loan types.
H3. Underwriting policy is associated with portfolio quality.	Compare policy-version outcomes while considering customer mix and cohort maturity.
H4. Acquisition channels differ in risk quality.	Compare channel-level rates, loan volumes and customer/loan mix.
H5. Portfolio growth and credit risk may move differently.	Compare application and loan growth with default, delinquency and write-off trends.
H6. Higher delinquency/default may increase financial pressure.	Analyze credit-loss trends and, where reliable, recovery and collection measures.

These hypotheses are exploratory, not assumptions that the analysis must confirm. Observed differences may be affected by customer selection, loan maturity, exposure, data quality and other factors.

5. Data Cleaning and Preparation in Python/Pandas

Python and Pandas were used for source-data inspection, basic cleaning and validation before detailed SQL analysis. The aim was to make the tables usable and to identify limitations, rather than force inconsistent source tables into artificial agreement.

1. Load the CSV files and inspect dimensions, column names, data types and sample records.
2. Review table grain, key uniqueness, missing identifiers and join relationships.
3. Identify missing values and distinguish true duplicate records from valid repeated events such as multiple repayments for one loan.
4. Standardize dates, numeric financial/risk fields, categorical values and flags.
5. Check business-rule validity, including implausible amounts, invalid dates and missing keys.
6. Compare repeated category fields before removing redundancy. For example, application_channel and acquisition_channel in the applications table were found to match; IDs required for joins were retained.
7. Investigate differences across loans, collections, repayments and risk tables; record unresolved discrepancies and define a consistent source for each metric.

Metric Definitions and Data-Quality Decisions

Metric / issue	Project decision
Default	Use loans.loan_status = 'Defaulted' as the agreed source of truth for the default metric.
Delinquency	Use collections delinquency information, including 30+, 60+ and 90+ DPD stages; count a loan once in loan-level measures.
Written-off loans	Use the agreed loan-status/write-off definition consistently; do not treat write-off as necessarily mutually exclusive with default.
Repayment vs collections conflicts	Some records disagree. Do not invent a reconciliation; document the limitation and use the agreed source for each metric.
risk_score direction	The field appeared inconsistent with risk-band direction and other indicators; do not rely on it for the main interpretation until validated.
Repeated event records	Respect one-to-many relationships to avoid inflating distinct loan counts.

The analysis is therefore based on documented metric definitions and caveats. The data should not be described as fully reconciled.

6. SQL Server Analysis

SQL Server was used to join the relevant tables, aggregate loan outcomes, calculate rates and compare segments. The SQL analysis was organized around the business questions rather than as unrelated query exercises.

Business question	SQL approach	Output
How large is the portfolio?	Count applications and distinct loans; aggregate statuses and disbursements.	Portfolio scale, composition and funnel context
How is portfolio quality changing?	Calculate default, delinquency and write-off counts/rates by period.	Risk trends and cohort comparisons
Which customer groups have higher risk?	Group distinct loan outcomes by customer attributes.	Customer-segment rates
Which loan characteristics are associated with risk?	Compare outcomes by loan type and underwriting policy.	Product/policy comparisons
Which channels are associated with higher risk?	Aggregate distinct loans, defaults and delinquent loans by acquisition channel.	Channel-level risk concentrations
Do risk indicators align with outcomes?	Join outcomes to credit score, PD, DTI and risk band using appropriate record selection.	Risk-segmentation validation
What is the financial impact?	Analyze monthly credit losses and available collection/recovery measures.	Financial impact and recovery context

Calculation Principles

- Default rate = distinct defaulted loans divided by the defined loan population in scope.
- Delinquency rate = distinct delinquent loans divided by the defined loan population in scope.
- Written-off rate = distinct written-off loans divided by the defined loan population in scope.
- Credit loss is a monetary outcome and should not be confused with a loan count or a default rate.

- Use consistent denominators and distinct loan IDs. Avoid multiplying counts through one-to-many joins.
- For cohort analysis, use disbursement year to define the loan cohort; newer cohorts may have shorter observation periods.

7. Key Findings

7.1 Portfolio scale and composition

Dashboard metric	Reported value
Total applications	120.25K
Total loans	77.03K
Delinquent loans	3,656
Defaulted loans	2,989
Written-off loans	903
Paid share of loan distribution	51.76%
Active share of loan distribution	39.79%
Defaulted share of loan distribution	3.88%

The portfolio contains paid, active, defaulted, restructured and written-off outcomes. Delinquency, default and write-off counts are separate indicators and may overlap; they should not be added together as if they represented unique affected loans.

7.2 Cohort default and write-off rates

Disbursement cohort	Default rate	Written-off rate
2024	4.33%	1.30%
2025	3.75%	1.13%

The 2025 cohort has lower observed default and write-off rates than the 2024 cohort, even though the number of defaulted loans increased. This can happen when the portfolio grows. The rate decline does not prove an improvement because newer loans may have had less time to default or be written off.

7.3 Acquisition-channel risk

Acquisition channel	Delinquency rate	Default rate
Social Media	8.78%	7.23%
Affiliate	6.40%	5.14%
Paid Search	6.20%	4.98%
Partner	4.51%	3.86%
Email	1.73%	1.32%

Social Media has the highest displayed rates among the listed channels, while Email has the lowest. Affiliate and Paid Search also show relatively elevated rates. The next step is to examine loan volumes, customer mix, average loan size and financial loss before making channel-spend decisions.

7.4 Customer employment status

Employment status	Delinquency rate	Default rate
Unemployed	5.5%	4.7%
Self-employed	4.8%	3.9%
Salaried	4.7%	3.9%
Contract	4.5%	3.6%

Unemployed customers have the highest displayed rates and contract employees the lowest among the four shown. Employment status is a useful segment for further affordability analysis, but should not be used alone to determine creditworthiness.

7.5 Loan type and underwriting policy

Loan type	Default rate	Delinquency rate
Consumer	3.9%	4.7%
Personal	3.9%	4.8%
Education	3.7%	4.5%
Emergency	3.7%	4.5%

Consumer and personal loans have slightly higher displayed rates than education and emergency loans. The differences are modest and should be interpreted alongside loan volume, customer mix and exposure.

Underwriting policy version	Observed default rate
V1	4.36%
V2	3.60%

Policy V2 has a lower observed default rate in the analyzed data. This is a signal for further evaluation, not proof that V2 caused better outcomes. Comparable cohort maturity, borrower profiles and selection criteria should be checked.

7.6 Credit loss and risk-band validation

The dashboard's credit-loss trend rises through much of the displayed period before falling in the latest month. Credit losses can move differently from default rates because portfolio exposure, loan size, loss severity and recoveries can change. The latest drop should be checked for data completeness and cohort maturity.

An earlier SQL analysis showed a counterintuitive result: the low-risk band had a higher observed default rate than the high- and very-high-risk bands. The recorded risk_score also appeared inconsistent with risk-band direction and other indicators. The project therefore treats risk-band reliability as an unresolved data-quality issue rather than assuming the score is valid.

8. Power BI Dashboard

The reporting layer contains two connected pages. Page 1 establishes portfolio performance; Page 2 explores risk segmentation and credit-loss trends.

Page 1 — Loan Portfolio Overview

- KPI cards: total applications, total loans, delinquent loans, defaulted loans and written-off loans.
- Loan-distribution donut chart: paid, active, defaulted, restructured and written-off statuses.
- Applications and loans over time.
- Delinquent, defaulted and written-off loans over time.
- Slicers: year, loan type and acquisition channel.

Purpose: give stakeholders a fast view of portfolio size, composition and the evolution of risk-related outcomes.

Page 2 — Delinquency and Default Drivers

- Risk by Customer Characteristics: selectable customer views, including income, job sector, gender and risk band, as configured in the report.
- Risk by Loan Characteristics: comparison by loan type or underwriting policy.
- Credit Loss Trend by Month.
- Risk by Acquisition Channel table with delinquency and default rates.
- Slicers: year and acquisition channel.

Purpose: let stakeholders move from a portfolio-level signal to the customer, loan or channel segments that warrant further investigation.

Stakeholder workflow

8. Review the portfolio KPIs and time trends.
9. Identify an outcome or period that needs attention.
10. Filter the relevant year or acquisition channel.

11. Compare customer characteristics, loan types and policy versions.
12. Validate segment size, cohort maturity and financial exposure.
13. Prioritize an investigation or operational action based on the evidence.

Collections and recovery were included in the broader analysis scope, but the supplied screenshots show two dashboard pages and do not show a dedicated collections/recovery page.

9. Business Recommendations

Investigate Social Media acquisition first

Recommended action: Compare channel risk rates with loan volume, average loan size, customer income, DTI, credit profile, campaign/partner and credit losses. Review campaign quality before reducing spend.

Monitoring measure: Monitor default and delinquency rates, risk-adjusted loss, conversion and profitable acquisition.

Strengthen affordability assessment

Recommended action: Investigate income stability, existing debt obligations and repayment capacity across employment segments. Evaluate combinations of attributes rather than using employment status alone.

Monitoring measure: Monitor subsequent delinquency by comparable customer/loan cohorts and avoid unnecessarily excluding creditworthy applicants.

Validate underwriting-policy performance

Recommended action: Compare V1 and V2 using similar borrower profiles, loan types, disbursement cohorts and observation periods. Consider both default frequency and loss severity.

Monitoring measure: Monitor mature-cohort default rates, loss outcomes, approvals and portfolio profitability.

Validate risk-score and risk-band logic

Recommended action: Confirm score direction, band assignment rules and PD calibration. Do not rely on the disputed risk_score field until its meaning and behavior are validated.

Monitoring measure: Check whether risk bands consistently differentiate observed outcomes in sufficiently mature cohorts.

Monitor losses alongside default rates

Recommended action: Track credit losses, write-offs, recoveries, collection costs and exposure. Investigate sharp monthly movements and verify that recent data is complete.

Monitoring measure: Measure loss relative to a clearly defined exposure base, recovery outcomes and adverse monthly movements.

10. Limitations and Responsible Interpretation

- Synthetic data: results demonstrate an analytical workflow and are not findings about an actual lender.
- Source inconsistencies: repayment and collection records do not always agree; metric definitions and unresolved discrepancies must be disclosed.
- Association is not causation: higher rates identify segments for investigation but do not prove that a channel, customer attribute or policy caused default.
- Cohort maturity: recent loans may not have had enough time to default or be written off, so time comparisons require care.
- Join and denominator risk: one-to-many joins can inflate counts; rates must use consistent populations and distinct loan IDs.
- Financial reconciliation: monthly business KPIs and loan-level aggregates may differ unless their definitions, dates and scopes are aligned.

11. Conclusion

LendingGuard demonstrates an end-to-end Data Analyst workflow: defining a lending-risk problem, validating a multi-table dataset, preparing data with Python/Pandas, investigating business questions in SQL Server and communicating results through a two-page Power BI dashboard.

The analysis highlights acquisition channels with elevated observed risk, differences across customer and loan segments, lower observed default and write-off rates for the 2025 cohort, a lower observed default rate under policy V2, a rising credit-loss trend over much of the period, and a risk-band inconsistency that needs validation. These findings support targeted investigation rather than immediate causal claims.

FINAL BUSINESS TAKEAWAY

A lender should not assess portfolio quality through default rate alone. It should understand risk concentrations, cohort maturity, exposure, credit losses and recovery outcomes together. The dashboard provides a decision-support view, while data-quality validation and deeper cohort analysis remain essential before operational changes are made.

Appendix A — Core Definitions

Term	Working definition in this project
Defaulted loan	Loan with loans.loan_status = 'Defaulted', as agreed for the main default metric.
Delinquent loan	Distinct loan identified from collections delinquency information at the defined DPD threshold/stage; metric scope must be stated.
Written-off loan	Loan classified as written off under the agreed source/status definition.
Default rate	Distinct defaulted loans divided by the defined loan population in scope.
Delinquency rate	Distinct delinquent loans divided by the defined loan population in scope.
Written-off rate	Distinct written-off loans divided by the defined loan population in scope.
Cohort year	Year of disbursement date for the loan-cohort comparison.
Credit loss	Monetary loss measure; definition should be checked against the source field/business KPI.